10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

# Chapter 11 — Transfer of Thermal Energy

Solved Exercise | New Book (PECTAA 2026)

## Exercise 11 (Solved)

This exercise covers the three ways in which heat travels — conduction, convection and radiation — the greenhouse effect and the everyday effects of each method.

### A. Multiple Choice Questions

**11.1 Which method of heat transfer requires direct contact between particles?**

(a) Convection
(b) Radiation
(c) Conduction
(d) Evaporation

**Answer: (c) Conduction** — in conduction, heat passes from particle to particle by direct contact; no particle travels from one place to another.

**11.2 What causes convection currents in liquids and gases?**

(a) The continuous movement of particles in all directions
(b) The rising of warm regions and sinking of cooler regions
(c) The conduction of heat through the medium
(d) The reflection of heat waves

**Answer: (b) The rising of warm regions and sinking of cooler regions** — the heated fluid becomes less dense and rises, the cooler fluid sinks to take its place, and this circulating flow is called a convection current.

**11.3 The natural phenomenon caused by convection currents is:**

(a) earthquakes
(b) land and sea breezes
(c) magnetic fields
(d) solar eclipses

**Answer: (b) land and sea breezes** — unequal heating of land and sea sets up convection currents in the air along the coast, giving sea breeze by day and land breeze by night.

**11.4 How does heat transfer occur through radiation?**

(a) Through direct contact between objects
(b) Through the movement of heated fluid
(c) Through electromagnetic waves without needing a medium
(d) Through the absorption of water molecules

**Answer: (c) Through electromagnetic waves without needing a medium** — radiation is the transfer of heat by electromagnetic (mainly infrared) waves, which travel even through a vacuum — that is how the Sun's heat reaches the Earth.

**11.5 The surface which absorbs and emits the most radiation is:**

(a) smooth and shiny surfaces
(b) dark and rough surfaces
(c) transparent and thin surfaces
(d) polished and white surfaces

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**Answer: (b) dark and rough surfaces** — a dull black rough surface is the best absorber and the best emitter; a shiny silver surface is the poorest.

### 11.6 What is the role of the greenhouse effect in Earth's temperature regulation?

(a) It allows all heat to escape into space
(b) It blocks heat from entering the atmosphere
(c) It traps heat to maintain a stable temperature
(d) It causes immediate cooling of Earth's surface

**Answer: (c) It traps heat to maintain a stable temperature** — the greenhouse gases allow the Sun's short-wave radiation in but absorb the outgoing long-wave radiation, so the Earth stays warm enough for life.

### 11.7 The Sun's heat reaches the Earth:

(a) through conduction between air particles
(b) through convection in space
(c) through radiation via electromagnetic waves
(d) through reflection by the moon

**Answer: (c) through radiation via electromagnetic waves** — the space between the Sun and the Earth has no medium, and only radiation can cross empty space.

## B. Short Answer Questions

### 11.1 What are the three primary types of heat transfer?

The three types are conduction, convection and radiation.

1. Conduction: heat passes through a solid from particle to particle by direct contact, without the particles themselves moving from place to place. For example, the handle of a metal spoon becomes hot when its other end is in tea.

2. Convection: heat is carried by the actual movement of a liquid or gas. The warm fluid rises and the cold fluid sinks, forming a convection current, as in water boiling in a pan or air circulating in a room.

3. Radiation: heat travels as infrared electromagnetic waves and needs no medium at all; that is how we feel the heat of a fire or of the Sun.

### 11.2 Why does heat travel faster in metals compared to non-metals?

Metals contain free electrons. These electrons are not fixed to any atom; they move freely throughout the metal and carry energy with them as they move. When one end of a metal rod is heated, the free electrons gain energy and travel quickly along the rod, handing the energy to other atoms on the way. So heat is carried both by the vibrations of the atoms and by these fast-moving electrons.

In non-metals such as wood, plastic, glass or cloth there are no free electrons. Heat can only pass on slowly by the vibration of atoms, therefore non-metals are poor conductors and are called insulators. That is why cooking pans are made of metal but their handles are made of wood or plastic.

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### 11.3 How does convection transfer heat in liquids and gases?

When a liquid or gas is heated from below, its particles gain energy, move faster and spread out, so the heated part becomes less dense. This lighter warm fluid rises upward. The cooler, denser fluid from above sinks down, gets heated in turn and rises. In this way a circulating flow is set up in the fluid, and this flow is called a convection current.

Heat is thus carried from the hot region to the cold region by the movement of the medium itself. Examples: water heating in a pan, the warming of a room by a heater placed near the floor, the draught in a chimney, and land and sea breezes.

### 11.4 What natural phenomenon causes land and sea breezes?

Convection currents in air cause land and sea breezes. Land gets hot and cold faster than water, because water has a much larger specific heat capacity.

During the day the land heats up quickly, the air above it becomes warm and rises, and the cool air from the sea blows towards the land to fill its place. This is the sea breeze.

At night the land cools quickly while the sea stays warmer, so the air above the sea rises and the cooler air from the land blows out towards the sea. This is the land breeze. In this way the breeze blows from sea to land in the day and from land to sea at night.

### 11.5 How do birds and gliders take advantage of convection currents?

When the ground is heated by the Sun, the air above it becomes warm, expands and rises in narrow columns called thermals. Birds such as eagles and gliders spread their wide wings and simply ride upward on these rising warm-air currents.

Because the rising air does the lifting, they can stay in the air for hours without flapping their wings and so save a lot of energy. When they reach a height where the air is cool, they glide forward and slowly downwards until they find another thermal. Glider pilots use exactly the same trick.

### 11.6 What is radiation, and how does it transfer heat?

Radiation is the transfer of heat in the form of electromagnetic waves, mainly infrared waves. Every object above absolute zero emits such waves, and the hotter the object, the more energy it radiates.

These waves need no material medium, so radiation can carry heat through a vacuum — that is how the Sun's heat reaches the Earth through empty space. When the waves strike another object, part of them is absorbed and the object warms up. That is why we feel warm while sitting in front of a heater or a fire even without touching it.

### 11.7 Why does a dull black surface emit and absorb more radiation than a shiny silver surface?

A dull black surface is a good absorber and a good emitter of radiation. The rough, dark surface traps the incident infrared radiation in its tiny irregularities and absorbs most of it, instead of reflecting it back. When such a surface is hot it also radiates energy away very effectively.

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A shiny silver surface is highly reflective: most of the radiation falling on it bounces back, so little is absorbed, and for the same reason it emits very little. That is why the absorbing plate of a solar water heater is painted black, whereas the bottle of a thermos flask is silvery to reduce heat loss by radiation.

### 11.8 How does the greenhouse effect help regulate Earth's temperature?

The Sun's radiation is short-wavelength, so the gases of the atmosphere let most of it pass through and the Earth's surface is warmed. The Earth then radiates energy back as long-wavelength infrared radiation. Greenhouse gases — mainly carbon dioxide, water vapour, methane and nitrous oxide — absorb this long-wave radiation and re-radiate part of it back towards the surface.

This trapped energy keeps the average temperature of the Earth near 15 °C instead of about −18 °C, which would be the case with no atmosphere. In this way the greenhouse effect keeps the Earth warm enough for life and prevents very large differences between day and night temperatures. The problem today is that extra greenhouse gases from burning fuels trap too much heat, causing global warming.

## C. Constructed Response Questions

### 11.1 During the Leslie's Cube experiment, why does the dull black surface radiate more heat than the shiny silver one, even though they are at the same temperature?

Leslie's cube is a metal box filled with hot water, with its four faces finished differently: dull black, shiny silver, white and rough. A radiation detector is held at the same distance from each face in turn.

All the faces are at the same temperature, yet the detector shows the largest reading in front of the dull black face and the smallest in front of the shiny silver face. The reason is the nature of the surface. The dull black rough surface has a high emissivity: it radiates infrared waves strongly. The shiny silver surface has a low emissivity: it reflects radiation back and emits very little.

The experiment proves two things — that the rate of emission depends on the colour and texture of the surface and not only on temperature, and that a good absorber is also a good emitter. This is why black surfaces are used where heat must be radiated away quickly, and shiny surfaces where heat loss must be prevented.

### 11.2 How does the size (surface area) of a radiator affect the amount of heat it emits through radiation? Explain using a real-world example.

The heat radiated in a given time is directly proportional to the surface area of the object. If the area is doubled, keeping the temperature the same, the object radiates roughly twice as much energy. A larger surface also lets more air touch the object, which adds to the heat given out by convection.

That is why a car radiator, a room radiator and a computer heat sink all have many thin fins: the fins increase the surface area without increasing the size of the object much, so the heat is lost quickly and the engine or the chip stays cool. In the same way tea cools faster in a wide saucer than in a tall cup, and a small cube of ice melts faster than a large block of the same mass.

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### 11.3 Why is convection not possible in solids, and how do convection currents in liquids and gases compensate for their poor thermal conductivity?

In a solid the particles are held in fixed positions by strong intermolecular forces. They can only vibrate about their mean positions; they cannot move from one place to another. Convection needs the heated material itself to travel and carry the heat with it, so convection cannot occur in a solid — heat can only pass through it by conduction.

Liquids and gases are poor conductors because their particles are far apart and have no free electrons, so conduction alone would be very slow. Convection makes up for this. The heated fluid expands, becomes less dense and rises, while the cooler fluid sinks; the whole fluid circulates in a convection current and carries heat quickly through the volume. Because of convection, a pan of water heats in a few minutes, whereas a solid of the same size would take much longer.

### 11.4 How do birds like eagles and gliders manage to stay aloft without flapping their wings for hours? Explain the scientific principle involved.

The principle is convection. The Sun heats the land, and the land heats the air in contact with it. This warm air expands, becomes less dense and rises in a column called a thermal. Where the ground is dark — a field, a road, a built-up area — the thermals are stronger.

An eagle or a glider spreads its broad wings to catch this rising air. The upward force of the rising thermal balances the weight of the bird, so it gains height without any effort from its wings. Once high up, it glides forward while slowly losing height, and then finds another thermal to rise again. By repeating this, a bird can stay in the sky for hours without flapping, and it uses very little energy.

### 11.5 Why does a hot metal object placed in a cooler room eventually cool down? Explain using the concepts of radiation and thermal equilibrium.

A hot object radiates infrared energy at a rate that depends on its temperature: the hotter it is, the more it radiates. The walls and the objects of the cooler room also radiate, but at a lower rate because they are cooler.

So, at the beginning, the hot object emits more energy than it absorbs, and its temperature falls. As it cools, the difference between the two rates becomes smaller and smaller. It also loses heat by convection to the surrounding air and by conduction to the table it rests on.

Eventually the object comes to the same temperature as its surroundings. Then it emits and absorbs energy at equal rates, no further change of temperature takes place, and the object and the room are in thermal equilibrium.

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## D. Comprehensive Questions

### 11.1 Describe how convection occurs in fluids. How do convection currents contribute to natural phenomena such as sea breezes and wind patterns?

Convection in fluids. When a liquid or gas is heated, the particles of the heated region gain kinetic energy and move farther apart, so that region becomes less dense and its volume increases. Being lighter, it rises. The colder, denser fluid above sinks down to be heated in turn. This continuous circulation of the heated fluid is called a convection current, and it carries the heat from one place to another with the medium itself. Convection needs gravity and a fluid; it cannot occur in a solid.

Sea breeze. Land heats up and cools down faster than water. In the daytime the land becomes much hotter than the sea, so the air above the land rises and the cooler air over the sea moves in towards the land — this daytime wind is the sea breeze. At night the land cools faster, the air above the sea becomes warmer and rises, and cool air flows from land to sea — the land breeze.

Wind patterns. The same unequal heating happens on a large scale. Near the equator the land and air are heated strongly, the warm air rises, and cooler air comes in from the higher latitudes; this sets up regular winds such as the trade winds and the monsoon. Warm air rising from the sea also forms clouds and rain. So all the winds of the world are, in the end, giant convection currents in the atmosphere.

### 11.2 How does radiation differ from conduction and convection in terms of heat transfer? Give examples of radiation in everyday life and explain how different surfaces affect radiation absorption and emission.

Radiation transfers heat by electromagnetic waves and does not need any medium, so it can travel through a vacuum. Conduction needs a solid medium in which particles are in contact, and convection needs a fluid that can circulate. Conduction and convection both involve matter moving or passing on energy, while radiation is the only method that reaches us across empty space from the Sun.

Everyday examples: feeling the warmth of a fire or a heater without touching it; the heat of the Sun reaching the Earth; a room being warmed by an electric heater's red-hot element; cooking bread in a toaster where the slice browns by radiation; a thermos flask keeping tea hot by using a silvered surface; and the sensation of cold when we sit near a cold window, because our body radiates heat towards it.

| Surface | Absorption | Emission | Example / use |
|---|---|---|---|
| **Dull black, rough** | absorbs most of the radiation falling on it | emits strongly when hot | black plate of a solar heater; radiator painted black |
| **White, light** | absorbs little, reflects much | emits little | white clothes in summer; white roofs |

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| Surface | Absorption | Emission | Example / use |
|---|---|---|---|
| **Shiny silver (polished)** | absorbs very little | emits very little | silvered walls of a thermos flask, survival blankets |
| **Transparent (glass)** | lets short waves through, absorbs long waves | — | glass of a greenhouse and of a car |

So the amount of radiation absorbed or emitted depends on the colour and texture of the surface: the darker and rougher the surface, the greater the absorption and emission; the shinier and lighter the surface, the smaller they are.

### 11.3 Discuss the role of the greenhouse effect in regulating Earth's temperature. What are the potential consequences of an enhanced greenhouse effect due to human activities?

The greenhouse effect is nature's way of keeping the Earth warm. Short-wave radiation from the Sun passes through the atmosphere and heats the land and the oceans. The Earth re-radiates this energy as long-wave infrared radiation. Gases such as carbon dioxide, water vapour, methane, nitrous oxide and ozone absorb the infrared radiation and send part of it back to the surface. As a result the average temperature of the Earth is about 15 °C instead of about −18 °C. Without this effect the Earth would be frozen and life would not be possible; the effect also keeps the difference between day and night temperatures small.

An enhanced greenhouse effect. Human activities — burning coal, oil and gas in factories, vehicles and power stations; cutting down forests; rice fields and cattle — have raised the amount of carbon dioxide and methane in the air. More greenhouse gas means more heat is trapped, so the Earth's average temperature is rising. This is global warming.

Its consequences are serious: melting of glaciers and polar ice, which raises the sea level and floods low coastal areas; more frequent and longer droughts and heat waves; heavier rainfall and floods in some regions; stronger storms; changes in the rainfall pattern that harm crops; the bleaching of coral reefs and the loss of many species; and spread of diseases such as malaria to new areas. The remedy is to reduce the burning of fossil fuels, plant more trees and use clean energy from the Sun and the wind.

### 11.4 Compare the absorption and emission of heat by different surfaces. How does the colour and texture of a surface affect its ability to absorb and emit heat?

Absorption and emission depend upon the colour and the texture of the surface, and the same surface that absorbs well also emits well.

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| Surface | Absorbs | Emits | Reason |
|---|---|---|---|
| **Dull black, rough** | most | most | the rough dark surface does not reflect the infrared waves; they are trapped and absorbed, and the same surface gives them out readily when hot |
| **White or light coloured** | least | least | light surfaces reflect most of the radiation, so little is absorbed and little is emitted |
| **Shiny, polished silver** | very little | very little | a mirror-like surface reflects almost all the radiation falling on it |
| **Rough compared with smooth** | more | more | a rough surface has a larger exposed area and repeated reflection inside its hollows, so more energy is caught |

Practical uses follow directly. Solar water heaters and solar cookers are painted black to absorb the maximum heat, and their covers are glass so that the long-wave radiation cannot escape. In hot countries houses are whitewashed and people wear white clothes, because a white surface reflects the Sun's radiation and keeps things cooler. A thermos flask is silvered inside to cut down both the absorption and the emission of radiation, and the vacuum between its double walls stops conduction and convection. Cooking pots are shiny, since a shiny surface absorbs less heat from the surroundings and keeps the food warm for longer.

### 11.5 Analyse how uneven heating of Earth's surface leads to the formation of convection currents and influence wind patterns.

The Earth's surface is not heated equally. Near the equator the Sun's rays fall almost vertically on a small area, so the ground and the air there become very hot. Near the poles the same rays fall at a slant and spread over a much larger area, so far less heat is received. Land heats faster than the sea, and dark ground faster than light ground. Hills, forests and deserts also heat differently.

Because of this uneven heating, warm air rises over the hot regions — over the equator, over the land in daytime, over a city rather than over a lake. Cooler, denser air then flows in from the sides to replace it, and the air that rose cools at a height and moves back towards the cooler region. This closed circulation is a convection cell, and on the Earth's surface it appears as wind.

In this way the global wind belts are produced: air rises near the equator and the surface air flows in from both sides as the trade winds; in the middle latitudes the westerlies blow; near the poles the cold air sinks and spreads outward. Local winds follow the same rule — sea and land breezes along the coast, mountain and valley breezes in hilly areas, and the monsoon, which is a seasonal reversal of the winds caused by the huge heating of the land in summer. Uneven heating is therefore the engine that drives all winds.

#### Points to Remember

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• Conduction: particle to particle, needs a medium, fastest in metals (free electrons).

• Convection: heat carried by the moving fluid; needs gravity; not possible in solids.

• Radiation: infrared electromagnetic waves; needs no medium; travels through vacuum.

• Dull black surface — best absorber and best emitter; shiny silver — poorest.

• Greenhouse gases ($CO_2$, $CH_4$, water vapour) trap long-wave radiation.

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